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## **Integrating Surface Well Testing Solutions to Enhance Real-Time Productivity in Underbalanced Coiled Tubing Drilling Operations: A Success Story in Onshore Sharjah, UAE**

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### **Abstract**

Underbalanced Coiled Tubing Drilling (UBCTD) operations are specifically designed to drill with controlled reservoir flow and are recognized for enhancing the well's performance and productivity. However, they require additional surface equipment for solids and hydrocarbon management, making the integration of Surface Well Testing (SWT) essential. This paper details how advancements in SWT were instrumental in improving real-time data acquisition and decision-making, optimizing productivity, and achieving minimal flaring during UBCTD operations while drilling 32,200 ft of open hole. The approach taken for an automated SWT design included the use of Programmable Logic Controller (PLC)-driven control safety systems and quantitative continuous solids measurement, along with precise gas reinjection and monitoring, to achieve complete operational control overproduced volumes, allowing for flaring or injection into the pipeline. The data sources were digitally integrated, enabling real-time monitoring of all parameters and facilitating faster, smarter decisions. Automation control and real-time access to data ensured continuous operations with zero downtime, as well as assurance of injected condensate and gas quantities. All data from the Coiled Tubing Unit (CTU), Drilling and Measurements (D&M), and SWT were digitally integrated. The automation and digital enablement strategy for the SWT equipment, integrated with CT parameters into the data acquisition system (DAQ), provided real-time feedback on choke positions, enhancing both efficiency and reliability during operations. To ensure uninterrupted operations, especially during production facility shutdowns, the traditional pneumatic system was upgraded to an automated gas diversion system, ensuring consistent performance throughout the entire operation. Furthermore, the real-time solid monitoring and advanced sample collection (even below 500g returns) enabled geologic control and real-time bio steering evaluation during drilling. This was achieved without using unnecessary water or chemicals for treatment. The gas reinjection module included automation of the gas diversion system, which allowed approximately 88% of the produced gas during drilling to be sent to the company's operating

gas plant. Only 12% of the produced gas was flared on-site, resulting in a direct reduction of over 26,7500 metric tons of carbon dioxide (CO<sub>2</sub>) throughout the drilling campaign. For the first time, this automated UBCTD SWT system was implemented in the UAE, enabling precise, real-time monitoring and adjustment of operational parameters at the surface. This resulted in substantial environmental benefits, including a significant reduction in CO<sub>2</sub> emissions through the optimization of gas injection strategies. Additionally, it allowed operations to proceed with enhanced safety and reliability, underscoring the viability of UBCTD technology in complex reservoir environments, which are common to the region.

## Introduction

### Coiled Tubing Under Balanced Drilling – Technology Overview

Conventional (overbalanced) drilling is the most common form of drilling but is known for generating impairment of the formation (skin) during the drilling process. Although interventions to reduce this damage are well known, there is no guarantee that the damage will be completely reversed or minimized. Underbalanced drilling (UBD) offers an alternative to mitigate formation damage during drilling by imposing a pressure differential that pushes formation fluids into the wellbore rather than allowing drilling fluid filtrate and solids to invade the formation pores. In conventional drilling, hydrostatic pressure is the primary barrier whereas in UBD the primary well control is based on flow and pressure control using specialized equipment and procedures; both techniques maintain the blow out prevention (BOP) as a secondary barrier. UBCTD has additional benefits when compared with Rig UBD, such as the ability to complete flow tests while drilling, the fact that CT is a continuous gas tight, allows for constant nitrogen (N<sub>2</sub>) circulation, can be equipped with e-line technology (allowing for continuous telemetry of downhole data), maintains personnel away from the wellhead during operations and, allows for snubbing and a smooth annulus pressure profile.

As with any technology, UBCTD presents both advantages and disadvantages, and there are both good and bad candidate wells for drilling using this method. Amongst the benefits of UBCTD, advantages, we can cite the reduced formation damage (and the associated increased well productivity), the reduction in lost circulation of drilling fluids into the formation (thus decreasing the chemicals utilization and its cost and environmental impact after well cleanup), the reduced – or absent – stimulation requirements and costs, an improved formation characterization and the absence of risking differential sticking and an enhanced drilling performance.

Despite its advantages in preserving reservoir productivity, UBCTD introduces significant technical and operational challenges. The continuous influx of formation fluids during drilling requires controlled and more sophisticated surface separation, measurement, and handling systems for gas, liquids, and solids. This complexity increases the requirements for surface facilities to manage and control the well's fluids. This management includes solid separation and measurement (to support geosteering decisions), liquid handling or disposal, and gas handling – either through flaring or reinjection in a production line, whenever one is available.

### Environmental considerations

In the UAE, federal legislation has regulated Flaring Hydrocarbons since 1999 ([Federal Law No. 24 of 1999](#)), which complements the *Petroleum Resources Conservation Law* (Law No. 8 of 1978, establishing that associated gas must primarily be reinjected into reservoirs or otherwise utilized for fuel or production operations). The government of the UAE have strengthened climate governance with Federal Decree-Law No. 11 of 2024, which requires further reductions in emissions. Additionally, the operator, a government-owned entity, has aligned with the national effort to achieve Net-Zero by 2050 and is committed to transparent reporting on greenhouse gas emissions.

## Surface Well Testing (SWT) and CTUBD operations

During UBD operations, it is expected to have volume changes minute to minute of multiphase compressible fluids with known pressure transients; the capability to handle large, frequent, and varied data streams is vital. At the surface, the designed separation equipment should be capable of metering all material flowing from the well, providing the operator with critical data. Analyses of the data enable an informed course of action to be taken, effecting a positive change in surface conditions while minimizing the impact on the exposed reservoir rock.

In this paper, the focus will be on the discussion of the surface systems implemented during a CTUBD campaign in the UAE, and how said setup integration allowed for improved real-time decision-making through automation of flow control and solids measurement, increased the productivity of the drilling operation, and led to a minimized gas flaring requirement, reducing the environmental impact of the entire campaign.

## Project Overview and Execution

### Project background

In 1981–1993, six wells were drilled in the Moveyeid field as producers. At its peak, the field, located in Emirate of Sharjah, UAE, produced natural gas at nearly 200 MMscf/day with associated condensate. The reservoir intervals are the Cretaceous-aged formations of the Thamama Group. By 2016, the reservoir pressure had dropped to nearly 5% from the initial conditions making the field heavily depleted. In January 2017, the operator conducted a pilot gas injection project in this field, injecting processed dry gas at 1,000 psi wellhead pressures. This pilot was successful in addressing key uncertainties, leading to the use of the Moveyeid field as a gas storage reservoir in the next phase of its development. Ever since, the field has been utilized as an underground gas storage facility, injecting and withdrawing processed natural gas from its four active wells.

Broadly, the operator's goal was to establish energy independence. Specifically, this meant that the gas storage facility would meet or exceed the Emirate's growing peak gas demands. In 2020, a feasibility study for UBCTD was undertaken to enhance the existing field wells, aiming to increase reservoir exposure and intercept fractures to boost injection and withdrawal capabilities. This increased well capacity would aim to provide sustainable gas rates for extended durations with the lowest possible cushion gas pressure and among other well stimulation options.

In 2024, a total of 21 laterals were drilled using UBCTD, spanning 32,244 ft across three wells, namely Moveyeid-03, Moveyeid-04, and Moveyeid-05. The Shuaiba and Kharaib formations within Thamama were the target, with specific high productivity biozones within these zones as preferred drilling zones [reference SPE-216458-MS]. The producer well demonstrated approximately a 500% increase in production and the two injector/producer wells showed a 40-50% gain in injection rates at 300 psi lower injection pressure. Additionally, the cushion gas requirements were reduced by ~25 BSCF and the project objectives were met [reference SPE-222915-MS].

During the drilling process, the liquid and gas rates, as well as the total solids produced. The solids were measured to determine the effectiveness of the hole cleaning process. The solids measurement was such that it allowed for supporting real-time trajectory optimization and steering decisions enabled by on-site bio-stratigraphical analysis of the cuttings [reference to SPE-222415-MS]<sup>10</sup>. In alignment with the health and safety environment, gas flaring was minimized.

Figure 1 illustrates a process flow diagram (PFD) of the surface system employed in the campaign, which enabled all program achievements. A more detailed discussion of the main modified components follows.

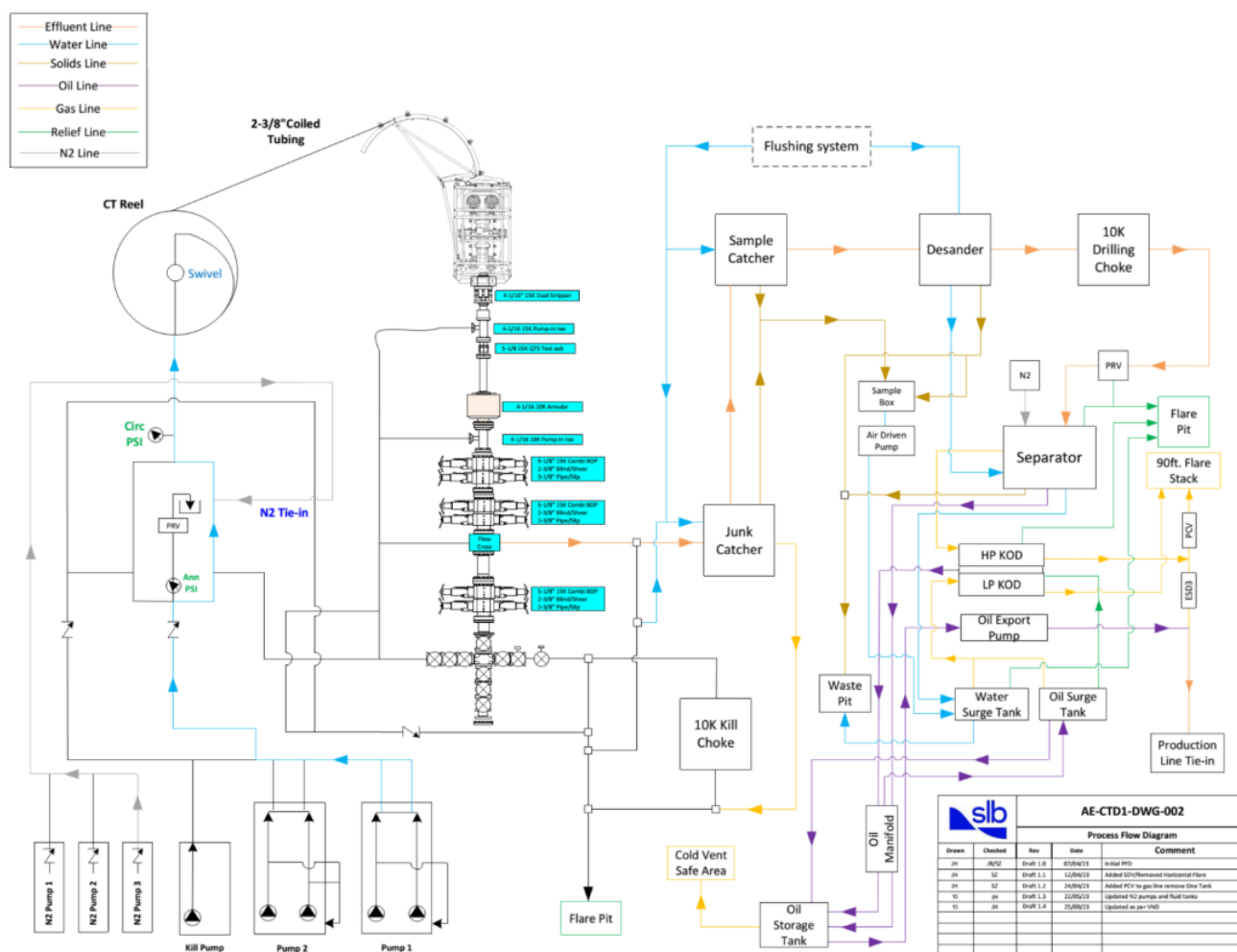


Figure 1—PFD of the surface system used in the campaign. It showcases the basic equipment required for the solids separation process, including the PLC-controlled ESD system (referred to as EESD in the drawing), the production line tie-in system, and the contingency flare.

## Automated Surface Well Testing (SWT) Design

To support the project's requirements, the SWT design was modified to accommodate the necessary components, enabling the achievement of the operator's goals. These can be divided into four main focuses: solid monitoring and management, digitalization of choke data, fluid conditioning, and automation of safety systems for continuous operations.

**Solids monitoring and management.** Solids management is a critical operation for UBD and well test design, as proper segregation of solids from fluids during drilling operations is paramount to avoid plugging or damaging the surface equipment, which can lead to the unplanned creation of an overbalance condition in the wellbore and ultimately affect the reservoir. Additionally, hole cleaning requires continuous monitoring of solids return, and bio steering involves analyzing the rock biostratigraphy properties to allow the continuation of the drilling.

To optimize solids segregation, a surface flow model was developed based on expected drilling parameters. The existing cyclonic solid separation system was sized to achieve maximum separation efficiency and to handle particles as small as 10  $\mu\text{m}$ , with a scheduled separation rate of 98%. To quantitatively assess the solids returned, two enclosed load cells with a sensitivity of 1kg were installed in the cyclonic separator. Continuous weight recording from the load cells enabled real-time measurement of the solids returned, synchronized with drilling logs. Because the two cells operated in parallel, solids collection could be diverted to one load cell while the other was isolated, allowing pressure to be equalized.

It was emptied to go on standby mode when a key player for the 1<sup>st</sup> stage solid separation system was the ability to collect and perform sampling on the solids return at the sample collection box, which enables the biostratigraphy engineer and geologist to analyze the sample in real time to ensure bio-steering and assure drilling can continue on a planned path. This setup allowed for continuous separation and sampling of formation solids during the drilling process.

**Programmable Logic Controller (PLC) controlled Emergency Shutdown (ESD) Systems.** The requirement to maintain both the integrity of the drilling conditions and the safety of personnel and the environment necessitated the installation of a PLC-controlled ESD, which can monitor input signals (such as pressure and temperature) and execute logic-based shutdowns to ensure continuous operations. Additionally, it allows for the integration and real-time monitoring through the surface data-acquisition system, permitting a controlled shut-in. Able to log events, interface with other systems if needed, and capable of managing multiple wells or zones in one logic system.

**Digitalized hydraulic choke manifolds.** The key to a successful CTUBD lies in maintaining a delicate balance between drill time efficiency and choke management. The choke valve is essential for controlling surface pressure and flow. Therefore, it was crucial for the operator and drilling engineer to determine the correct choke setting to enable drilling ahead while maintaining underbalanced conditions. Monitoring the actual position of the choke valve during operations was critical, which was achieved by converting the choke valve's variable voltage output to milliamps. Our engineers then digitalized this output through a PLC into a readable HART (Highway Addressable Remote Transducer) measurement (0% to 100%), which was integrated into the data acquisition system. This setup enhanced real-time monitoring, allowing on-the-spot decisions that improved both safety and operational efficiency.

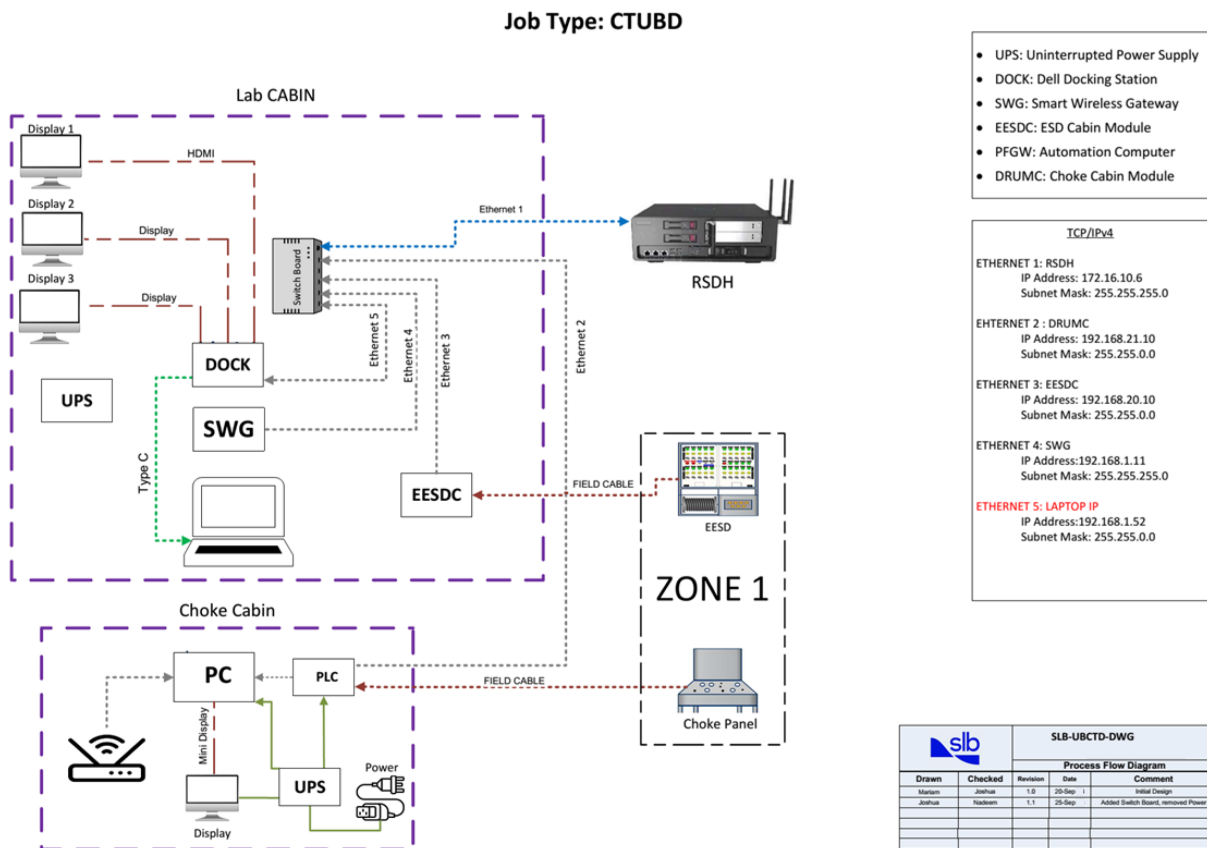


Figure 2—Showcase the internal network diagram done to achieve the digital data into one single data stream.



**Semi-Automatic Production Diverter to reduce CO2 footprint.** Part of the key to the project's success was to minimize flaring as low as reasonably achievable and allow gas to be exported to production while operations were ongoing, thereby reducing the impact of cutting gas production to Sharjah's gas plant. This plan was in alignment with the SERVICE COMPANY and the OPERATOR's Safety and Environment goals to reduce the CO2 footprint. For the SERVICE COMPANY to achieve this goal, various level of studies was performed:

- **Gas re-injection module and diversion system:** The key factor here was to ensure that the safety matrix and PLC were synchronized to ensure that all systems controlled by Electrical Emergency Shut down (EESD) were acting as intended to reduce human intervention. Once the operations team established the safety matrix, the engineering team proceeded to configure the PLC and integrate it into the process design, aligning with process plant safety concerns and limitations. The pressure switches and closed-loop system enabled the controlled export of gas, or, in the event of an emergency, the effective diversion to flare without interruption of operations, as shown in Figure 3.
- **Zero flaring strategy and semi-automation:** While 88% of the gas produced was re-injected (export to gas plant), the other 12% was flared, during emergencies or systems upsets. To achieve this, the primary task was to develop a relatively cost-effective solution for either reinjection or flaring. This was achieved by simply installing a PCV (Pressure control valve) and incorporating a pressure transducer from the gas exporting line, which senses changes in pressure to open partially or fully, allowing for continuous operations. The primary consideration here is the sizing studies conducted to enable an effective response, minimize time, and minimize human intervention.
- **Digital integration of CTU, D&M, and SWT into ONE:** One of the aims for this project was to be able to combine data from several business lines into one single data stream, where anyone could see a single data stream and make real-time decisions. The key to the success of this integration was the addition of a localized on-site server in conjunction with 4G wireless connectivity to ensure there was no data lag throughout the entire project. This innovation and ease into operations was possible by the implementation of RSDH (Remote System Data Hub) at the wellsite to ensure all data streams are transmitted into one single point, which transmitted data to a secure cloud data visualization platform in real time and allowed users to set up their visualization screens in any way for interpretation, monitoring, and operating.

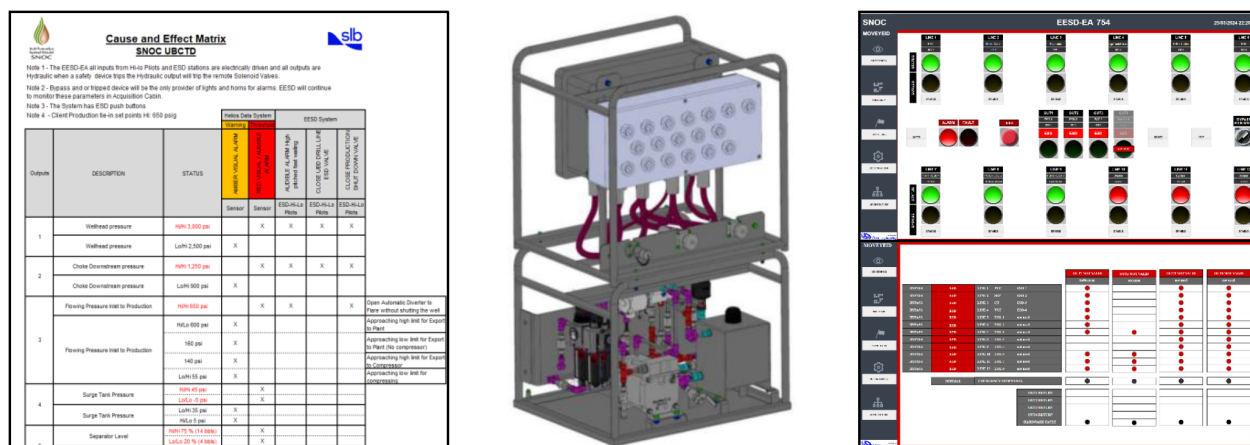


Figure 3—Safety Matrix and PLC safety system

## Results and Observations

Throughout the 219,204.00 person-hours of operation, the integration between CTD and SWT, facilitated through the same acquisition systems, allowed for close monitoring of surface pressure and flowing

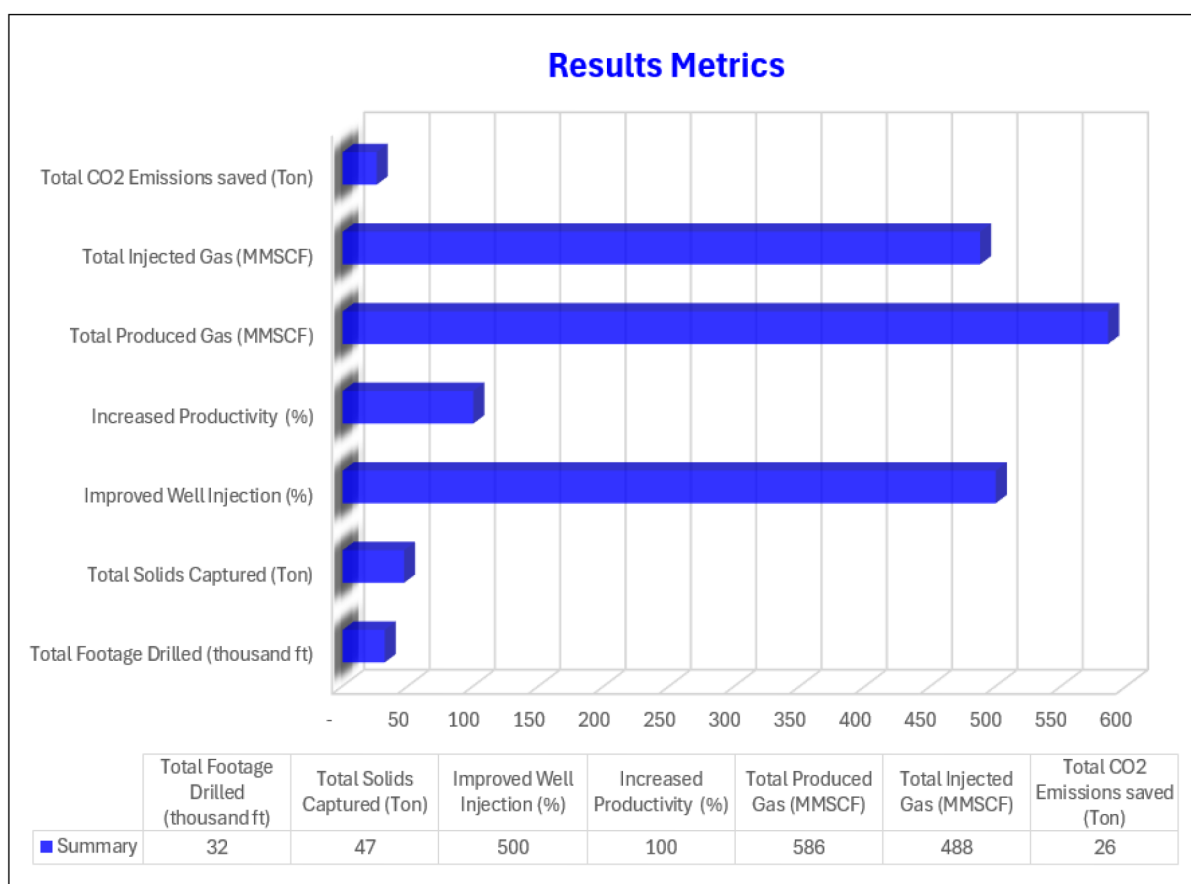
conditions in parallel with downhole conditions while drilling. That, together with the quantitative measurement of the solids returned, and the analysis of the said solids, enabled real-time feedback on the operation quality and control. In total, 47,700 kg of solids were produced, and over 200 solids analyses were performed. This led to strict Solids and geological control, resulting in the overall success of the project.

Reliability and efficiency gains were achieved by significantly increasing injectivity (doubling injection rates), and the surface injection pressures were reduced by approximately 20%. Lower pressure requirements directly led to lower compressor fuel consumption and a significant reduction in CO<sub>2</sub>e emissions, aligning with the OPERATOR's and the SERVICE COMPANY's sustainability goals.

Improvement in gas recovery post-CTUBD resulted in a 100% increase in well injection and a 500% increase in productivity. Proven the commercial viability of the Gas Injection Project to cover peak demand: The UBCTD campaign reduced cushion gas requirements by 25%-30% for the target of 50 MMscf over six months of production.

From the 586 MMscf of total gas produced, over 488 MMscf, or 88%, was reinjected, and only 12% was flared. This was made possible by the gas export and diversion system, which sent gas to production, resulting in a reduction of over 26.7 metric tons of CO<sub>2</sub>e in the atmosphere.

Operational Continuity: Zero surface-related downtime despite facility shutdowns (semi-automated diversion system).



**Figure 4—Showcase a high-level quantified summary of the achievements for the project.**

## Discussion

### Comparison to Previous Approaches

Traditional Surface Well Testing (SWT) setups are typically designed to measure productivity. Still, they often lack more advanced capabilities, such as four-phase separation, solids sampling, and integrated digital safety systems. In this project, those limitations were addressed through a redesigned SWT tailored for Underbalanced Drilling (UBD) conditions. By taking a more comprehensive approach—focusing on solids handling, fluid conditioning, and real-time data—the setup supported both the operational goals and the strict export requirements of the gas plant, while also aligning with the OPERATOR's and the SERVICE COMPANY's zero-flaring targets.

The semi-automated SWT system provided the team with greater control during drilling, particularly in monitoring flowback and solids return. With real-time data available from both surface and downhole sources, decisions can be made quickly and confidently. This played a crucial role in enhancing drilling performance and achieving extended lateral targets. The ability to sample and analyze solids on site meant that bio-steering could be guided with a higher degree of accuracy, helping ensure the well stayed on its planned path and delivered as expected.

From a safety and operations standpoint, the integration of a PLC-based Emergency Shutdown System (EESD) made a real difference. It enabled fast, automatic reactions to abnormal conditions, reduced the need for manual interventions, and gave teams greater confidence to operate continuously. These improvements supported a broader framework aimed at minimizing risk, encouraging teamwork, and ensuring that safety and environmental performance were built into every phase of the operation.

## Conclusions

The campaign conducted at Sharjah's underground gas storage facility marks a significant milestone in our approach to drilling and production in complex carbonate reservoirs. By combining underbalanced coiled tubing drilling (UBCTD) with an upgraded surface well testing (SWT) system, the project successfully re-entered and sidetracked three deviated wells, drilling over 32,200 feet of open hole laterals. This approach allowed the team to maximize reservoir contact while minimizing formation damage—both essential for a gas storage application where injectivity and deliverability are key. One of the standout results was the clear improvement in well performance. Injection rates doubled, and overall productivity increased by 500% following the campaign. The reservoir responded well to the lateral drilling, and surface injection pressures dropped by approximately 20%, resulting in compressors consuming less fuel—a small operational win that contributed to a much larger environmental benefit.

From a sustainability standpoint, this project pushed the boundaries of what's usually expected. Using a semi-automated gas diversion setup, we were able to export 88% of the produced gas back into the system during normal operations, with only 12% flared during upset conditions. That simple but effective design decision helped us avoid over 26.7 metric tons of CO<sub>2</sub> equivalent emissions. It also meant we kept gas flowing to the Sharjah plant throughout the work—no interruptions, no downtime.

Real-time solids monitoring and bio-steering also played a significant role. With over 47 metric tons of cuttings analyzed on site, we were able to stay on the planned drilling path and respond quickly to formation changes. Having live data from solids, downhole tools, and surface sensors all feeding into one system made it easier for the team to make informed decisions on the spot—whether that was adjusting choke settings, reacting to pressure changes, or verifying stratigraphic targets.

What made this campaign work wasn't just the technology; it was the integration across teams. Drilling, well testing, geology, safety, and digitalization all came together around a shared goal: completing the job efficiently, safely, and with the lowest possible environmental impact. The lessons learned here—especially those related to solids control, gas diversion, and data integration—are highly relevant for other operators



seeking to enhance performance while meeting environmental targets. In short, this project proved that it's possible to push technical boundaries without compromising on sustainability or safety. It sets a strong precedent for future gas storage developments in similar environments.

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